



SOFTWARE ENGINEERING Lab- CS353 Program-1

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Engineering*

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Experiment-1

1. Study the Agile process of software development

- ❖ Scrum
- ❖ XP
- ❖ Lean software development
- ❖ TDD
- ❖ Kanban

Agile Process Model

Scrum:

- Scrum principles are consistent with the agile platform and are used to guide development activities within a process that incorporates the following framework activities:
 - ✓ Requirements,
 - ✓ Analysis,
 - ✓ Design,
 - ✓ Evolution, and
 - ✓ Delivery.

Agile Process Model

Scrum:

- Within each framework activity, work tasks occur within a process pattern called a sprint.
- The work conducted within a sprint is adapted to the problem and is defined and often modified in real time by the Scrum team.
- The number of sprints required for each framework activity will vary depending on product complexity and size.

Agile Process Model

Scrum:

- Scrum emphasizes the use of a set of software process patterns that have proven effective for projects with tight timelines, changing requirements, and business criticality.
- Each of these process patterns defines a set of development actions:
 - ✓ Backlog
 - ✓ Sprints
 - ✓ Scrum meetings
 - ✓ Demos

Agile Process Model

Scrum:

Backlog:

- It is a prioritized list of project requirements or features that provide business value for the customer.
- Items can be added to the backlog at any time .
- The product manager assesses the backlog and updates priorities as required.

Agile Process Model

Scrum:

Sprint:

- It consists of work units that are required to achieve a requirement defined in the backlog that must be fit into a predefined time-box (typically 30 days).
- Changes (e.g., backlog work items) are not introduced during the sprint. Hence, the sprint allows team members to work in a short-term, but stable environment.

Agile Process Model

Scrum:

Scrum Meetings:

Scrum Meetings are (typically 15 minutes) meetings held daily by the Scrum team.

Three key questions are asked and answered by all team members:

- What did you do since the last team meeting?
- What obstacles are you encountering?
- What do you plan to accomplish by the next team meeting?

Agile Process Model

Scrum:

Scrum Meetings:

- A team leader, called a Scrum master, leads the meeting and assesses the responses from each person.
- The Scrum meeting helps the team to uncover potential problems as early as possible.

Agile Process Model

Scrum:

Demos:

- Demos deliver the software increment to the customer so that functionality that has been implemented can be demonstrated and evaluated by the customer.
- It is important to note that the demo may not contain all planned functionality, but rather those functions that can be delivered within the time-box that was established.

Agile Process Model

Members of Scrum Team:

- Product Owner,
- Scrum Master, and
- The Development Team.

Agile Process Model

Product owners:

- They are the champions for their product. They are focused on understanding business, customer, and market requirements.
- This role prioritizes the work to be done by the engineering team accordingly.

Agile Process Model

Effective product owners:

- ✓ Builds and manages the product backlog.
- ✓ Uses sprint backlog management by developing, ordering, and transparently managing the product backlog for effective planning and prioritization.
- ✓ Gives the team clear guidance on which features to deliver next.
- ✓ Decides when to ship the product with a predisposition toward more frequent delivery.

Agile Process Model

Product owners:

- A sprint backlog is a prioritized list of tasks selected from the product backlog for completion in a sprint.
- A product backlog is a prioritized list of work derived from the roadmap, guiding development teams on what to deliver next.

Agile Process Model

The Scrum master:

- Scrum masters are the champions of Scrum within their teams. They coach teams, product owners, and the business on the Scrum process and look for ways to fine-tune their practice.
- A key responsibility of the Scrum master is to enhance the Scrum team's effectiveness by coaching, removing impediments, and facilitating Scrum processes. This helps them improve overall team performance and delivery.

Agile Process Model

The Scrum Development Team:

- The Scrum Development Team is a self-organizing, cross-functional group responsible for delivering a potentially shippable product increment at the end of each sprint.
- They play a pivotal role in the success of Scrum by collaborating to meet sprint goals and ensuring high-quality deliverables.

Jira

- Jira is a, AI-powered **project management** and **issue-tracking tool** developed by Atlassian.
- It is widely used by software development, marketing, and HR teams to **plan, track, and release projects using Agile frameworks** like Scrum and Kanban.

Jira

How is JIRA used in Agile?

Step 1: Create a Project

- ☐ Choose Scrum or Kanban
- ☐ Define team members

Step 2: Create Issues

- Issues can be:
- ☐ Story (feature request)
- ☐ Task (work to do)
- ☐ Bug (problem to fix)

Jira

How is JIRA used in Agile? (Step-by-Step)

Step 3: Sprint Planning

- Select issues for the sprint
- Estimate effort

Step 4: Work & Update

- Developers move tasks:

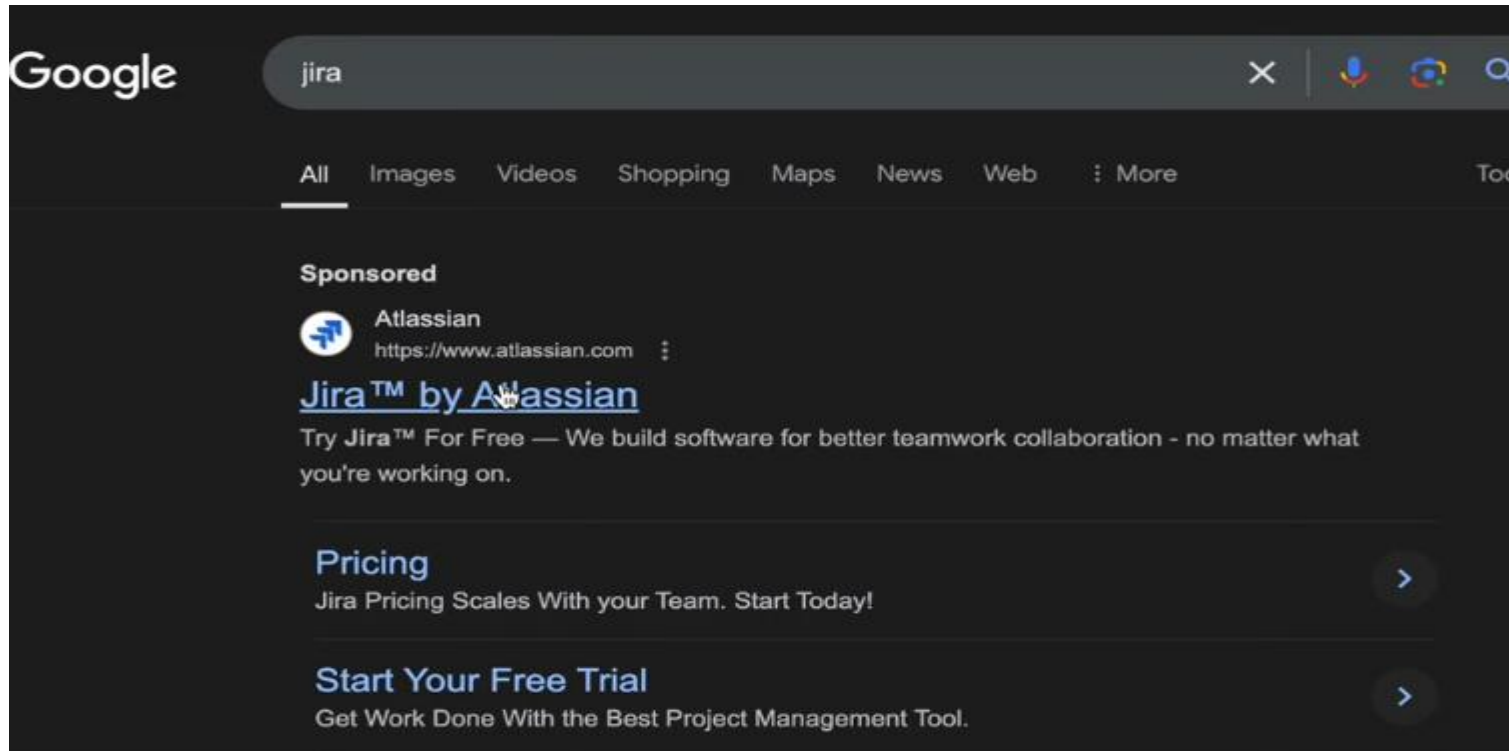
To Do → In Progress → Done

Step 5: Review Progress

Use boards and charts

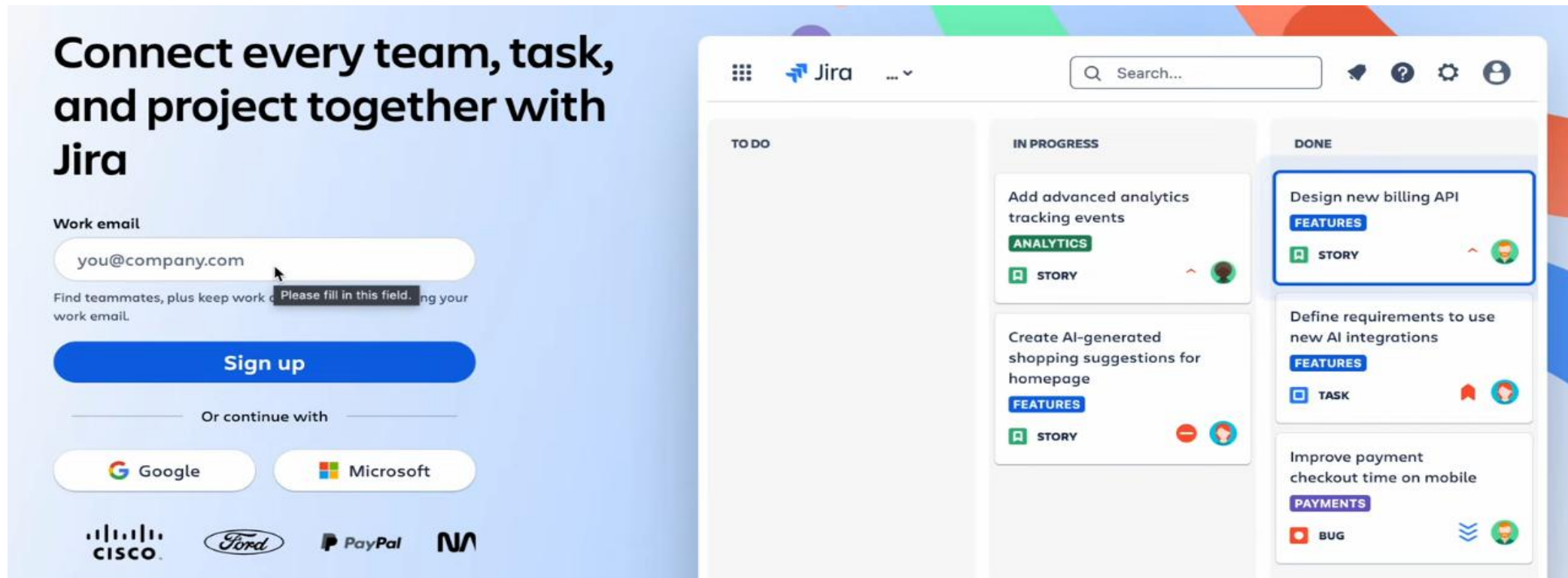
Creating Jira Account

Step-1: Go to the link <https://www.atlassian.com/try/cloud/signup>



Creating Jira Account

Step-2: Give your official college id for sign up



Creating Jira Account

Step-3: Open the mail and type the code which you have received and click verify.



We've emailed you a code

To complete your account setup, enter the code we've sent to:

mifeku@teleg.eu

Verify

[Didn't receive an email? Resend email](#)

Creating Jira Account

Step-4: Give full Name and Password and click Continue.



Email address verified 

Add your account details

Email address
mifeku@teleg.eu

Full name

Password
 

By signing up, I accept the Atlassian Cloud [Terms of Service](#) and acknowledge the [Privacy Policy](#).

Continue

This site is protected by reCAPTCHA and the Google [Privacy Policy](#) and [Terms of Service](#) apply.

Creating Jira Account

Step-5: Give site name for the Atlassian site and click continue.



Let's name your site

Your site name is part of your Jira URL. Most people use their team or company name.

Your site

.atlassian.net

This site name is just a suggestion. Feel free to change to something your team will recognize.

 Continue

Creating Jira Account

Step-5: Login to your account and Select Scrum from the template and



Welcome back, anusha.ise

It's free for up to 10 users - no credit card needed.

Your site

cambridge-team-qi875l2c

.atlassian.net



Continue

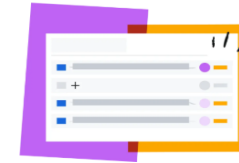
Select a template to get started

You can always change this later. Selecting a template won't limit what you can do.



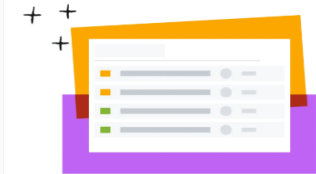
Kanban

Visualize work and maximize efficiency with a kanban board



Scrum

Plan, prioritize, and schedule sprints using scrum framework



Project management

Manage and track agile work plus integrate developer tools like GitHub

Continue

Extreme Programming (XP)

❖ **Focuses on engineering practices**

Key Practices:

☐ **Pair Programming**

☐ **Continuous Integration**

☐ **Refactoring**

☐ **Simple Design**

❖ **Emphasizes high code quality and customer involvement**

Lean Software Development:

❖ Inspired by Lean Manufacturing

Core Principles:

- ☐ Eliminate waste
- ☐ Build quality in
- ☐ Deliver fast
- ☐ Respect people
- ☐ Optimize the whole
- ❖ Focuses on efficiency and value delivery

Test-Driven Development (TDD)

- ❖ Development is driven by tests
- ❖ Follows Red – Green – Refactor cycle:
 - ☐ Write a failing test (Red)
 - ☐ Write minimum code to pass the test (Green)
 - ☐ Improve code quality (Refactor)
- ❖ Ensures reliable and maintainable code

Kanban

- ❖ **Visual workflow management method**
- ❖ **Uses a Kanban board with columns:**
 - ☐ **To Do list**
 - ☐ **In Progress**
 - ☐ **Testing**
 - ☐ **Done**
- ❖ **Limits Work in Progress (WIP)**
- ❖ **Suitable for continuous delivery systems**

Thank you